

Chintan Shah

Mechanical & Manufacturing Engineer

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SUMMARY

Mechanical & Manufacturing Engineer with 4+ years of experience spanning product development, robotic systems, manufacturing engineering, quality inspection, and design validation. Experienced in developing mechanical components and end-effectors, production test stations, and inspection processes from concept through validation and production release. Proficient in SolidWorks, CATIA, GD&T, tolerance stack-up, injection molding, CNC machining, prototyping, MSA, Gage R&R, DFMEA, and NPI. Strong background in cross-functional engineering, manufacturing documentation, test method validation, and design-for-manufacturability, with a focus on reliable, production-ready solutions.

WORK EXPERIENCE

Matic Robots, USA | Manufacturing Engineer

Jun 2025 - Current

- Led the development and production deployment of 3 Outgoing Quality Control (OQC) stations for airflow and system-performance checks, applying Measurement System Analysis (MSA) to maintain 150 units/day throughput without creating inspection bottlenecks.
- Established an in-line Track Testing process with a dual-surface test station, enabling 100% functional verification of cleaning performance under representative operating conditions prior to product release.
- Implemented a leak-integrity test method using regulated-pressure water immersion to identify pinhole and seal defects, replacing manual inspection with a standardized 30-second verification cycle.
- Developed a self-contained camera and microphone validation enclosure to isolate measurements from the closed-room process, improving test repeatability while targeting 50% cycle-time reduction and 60% lower floor-space requirements.
- Qualified inspection processes through Gage R&R and MSA using 3 operators × 3 trials, establishing repeatable and operator-independent measurement systems for production release and Test Method Validation.
- Standardized manufacturing documentation for 4 QC stations by creating SolidWorks models, engineering drawings, acceptance criteria, and work instructions, partnering with Engineering, Quality, and Production teams throughout NPI.

Meril Life Sciences Pvt. Ltd., India | Mechanical Design Engineer

Jan 2021 - Jul 2023

- Developed mechanical architecture for a six-axis robotic arm, detailing joint housings, arm segments, wrist flange, and end-effector interfaces while accommodating servo joints, encoders, brakes, and associated assembly requirements.
- Optimized a milling end-effector consisting of spindle mounting, tool coupling, and burr-holder components, addressing stiffness and vibration behavior to support precise resection performance.
- Integrated navigation hardware into the robotic assembly by designing mounting interfaces and cable-routing features for optical-tracker arrays and registration probes in coordination with Electronics Engineering.
- Produced SolidWorks and CATIA CAD models, assemblies, manufacturing drawings, and BOMs, applying GD&T and tolerance stack-up analysis to control critical interfaces and support manufacturing requirements.
- Developed injection-molded ABS, PC, and PC-ABS housings with appropriate wall thicknesses, draft angles, ribs, bosses, snap-fits, and gate locations, balancing manufacturability, thermal performance, and serviceability.
- Iterated mechanical prototypes using 3D printing, CNC machining, and sheet-metal fabrication, contributing to Design Verification (DV), Design Validation (DVT), DFMEA, and risk-management activities through successive design releases.

SKILLS

Mechanical Design & Product Development: Mechanical Design, Product Development, Robotic Systems, End-Effector Design, Design for Manufacturability (DFM), Design for Assembly (DFA), Concept Development, Design Iteration, Mechanical Integration

CAD & Engineering Documentation: SolidWorks, CATIA, 3D CAD Modeling, Assembly Design, Engineering Drawings, Bill of Materials (BOM), GD&T, Tolerance Stack-Up Analysis

Manufacturing & Prototyping: Injection Molding, CNC Machining, Sheet Metal Fabrication, 3D Printing, Prototyping, Production Release, Manufacturing Documentation, Standard Work

Quality, Testing & Validation: Measurement System Analysis (MSA), Gage R&R, Test Method Validation, Functional Testing, Leak Testing, Inspection Systems, Design Verification (DV), Design Validation (DVT), Acceptance Criteria

Engineering Analysis & Risk: DFMEA, Risk Management, Tolerance Analysis, Vibration Control, Structural Rigidity, Thermal Management, Root Cause Analysis

Manufacturing Engineering: New Product Introduction (NPI), Process Development, Quality Control, Production Throughput, Cycle-Time Improvement, Process Validation, Cross-Functional Collaboration

CERTIFICATION & ACHIEVEMENTS

ASQ Certified Quality Engineer (CQE).

SME Certified Manufacturing Engineer (CMfgE).

Patent Pending Project in USA (Project Name: Universal Syringe Adapter).

EDUCATION

Master's in Biomedical Engineering | Arizona State University, Tempe, Arizona